

Who Solves ARC Like a Human?

A Psychometric Comparison of LLM and Non-LLM Systems

@notcomplex_

April 2026

Abstract

This note asks a narrow question: once we add a few bona fide non-LLM ARC systems to the repository, do they look more human-like than the LLMs, less human-like, or simply different in a complementary way? The answer is mixed but fairly clear. On the ARC-2 human-overlap subset, the LLM aggregate sits comfortably inside the human split-half band, while the currently stored non-LLM systems do not. At the same time, the non-LLM systems are not just noise. A mid-training TRM checkpoint and the VARC predictions show small but real human-relevant structure, and the TRM+VARC union rescues a handful of human-easy / LLM-hard items above chance. The most important methodological point is that the low raw correlations are not automatically disappointing: the human benchmark itself is noisy, and low-score binary systems have a very wide same-accuracy random-placement null. The right comparison is therefore not raw correlation alone, but correlation relative to both human measurement noise and fixed-accuracy random baselines.

1 Question

The practical question here is not “are these systems literally the same as humans?” That is too vague to test. What we can test is whether their *item-difficulty profiles* line up with human solve rates in the same way.

The working nulls in this note are:

1. **Human-equivalence null:** the system’s item profile is exchangeable with a random human split-half profile.
2. **Feature-only null:** any observed human alignment is fully explained by easy task features such as size, color counts, or number of train pairs.
3. **No-extra-signal null:** after controlling for the LLM average, the non-LLM system has no leftover human-like signal.
4. **Fixed-accuracy null:** if a system solves exactly this many items, would a random placement of those wins look just as human-aligned?
5. **No-complementarity null:** non-LLMs do not rescue human-easy / LLM-hard items above chance.

Short version. The LLM aggregate is currently closer to the human ARC-2 difficulty axis than the non-LLM systems are. But the non-LLM systems are not useless or

purely random; they add some complementary signal, and one TRM checkpoint looks noticeably more human-like than later, higher-scoring checkpoints.

2 Data and Comparison Space

The main comparison space is ARC-2 Public Eval, because that is where the current human attempt dataset and the stored model outputs overlap cleanly.

- **Primary ARC-2 subset:** 110 test pairs with at least 8 human attempts each.
- **Human benchmark:** item-level human solve rates on those 110 pairs.
- **LLM reference:** the stored ARC-2 public-eval LLM prediction corpus already used in the main psychometric analysis.
- **Non-LLM systems:** TRM ARC-AGI-II checkpoints and VARC ARC-2 predictions.
- **ARC-1 sidecar:** CompressARC, evaluated only on the ARC-1 overlap that reappears inside the ARC-AGI-2 Public Train human file.

The main ARC-2 system profiles used below are:

- **LLM average:** the mean item-success profile across the stored ARC-2 LLMs.
- **Best-aligned LLM:** claude-opus-4-5-20251101-thinking-16k.
- **Best-score LLM:** gpt-5-2-2025-12-11-thinking-xhigh.
- **TRM mid-training:** TRM 361957 pass@2.
- **TRM best-score:** TRM 651522 pass@2.
- **VARC best ARC-2 profile:** VARC ARC-2_ViT pass@2.

3 How To Read The Correlations

The low raw correlations look underwhelming at first glance. They are less underwhelming once we put them in the right context.

3.1 Human reliability is only moderate

On the ARC-2 primary subset, the human split-half correlation has:

- median Pearson $r = 0.398$,
- 95% interval $[0.275, 0.515]$,
- implied full-length reliability $= 0.569$,
- approximate raw-correlation ceiling $= 0.755$.

So a raw item-level correlation around 0.3 or 0.4 is not small in the way it would be on a very clean human test. The benchmark itself is noisy.

3.2 Low-score systems have a wide null band

For low-accuracy binary systems, the same-accuracy random-placement null is wide. On ARC-2 it spans roughly $[-0.193, 0.191]$ for the systems studied here. That means a raw correlation of 0.11 may be indistinguishable from random placement at that score level, whereas a raw correlation of 0.21 can already be meaningful.

This is the main reason I do not want to over-index on the raw correlations alone.

4 Main ARC-2 Result

Figure 1 is the cleanest first look. It places each selected system’s correlation with human solve rates next to the human split-half band.

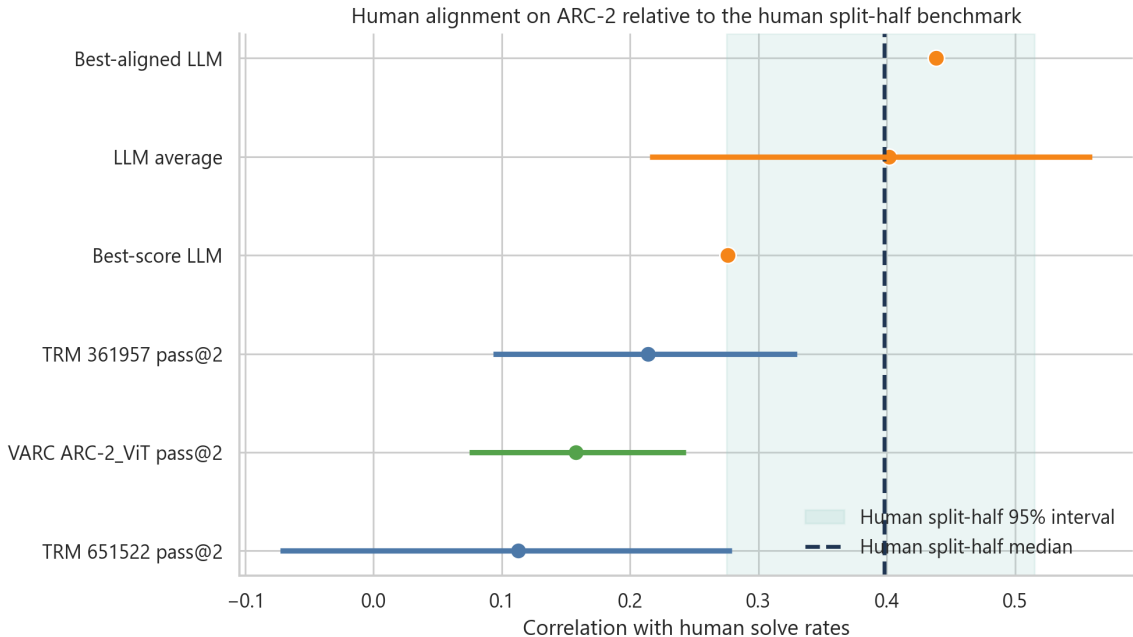


Figure 1: Selected ARC-2 system profiles compared against the human split-half benchmark. The shaded band is the human split-half 95% interval; the dashed line is the human split-half median.

The basic picture is simple:

- The **LLM average** lands inside the human split-half band ($r = 0.402$, $H1\ p = 0.950$).
- The **best-aligned LLM** also lands inside it ($r = 0.439$, $H1\ p = 0.506$).
- The **best-score LLM** is borderline rather than cleanly outside it ($r = 0.276$, $H1\ p = 0.052$).
- The three main **non-LLM profiles** are below the band and reject the human-equivalence null.

Table 1 gives the headline numbers I would actually want in front of me while reading this.

Table 1: Headline ARC-2 results on the 110-pair primary subset. H1 tests human-equivalence against the human split-half distribution. The fixed-accuracy null asks whether the observed human correlation is stronger than a random placement of the same number of wins. H3 tests for residual human signal after controlling for the LLM average.

Profile	Pair Acc.	Human r	H1 p	Fixed-Acc. p / H3 p
LLM average	0.108	0.402	0.950	—
Best-aligned LLM (Claude Opus 16k)	0.164	0.439	0.506	0.000 / —
Best-score LLM (GPT-5.2 xhigh)	0.491	0.276	0.052	0.003 / —
TRM 361957 pass@2	0.045	0.214	0.004	0.012 / 0.009
TRM 651522 pass@2	0.109	0.113	0.0004	0.122 / 0.105
VARC ARC-2_ViT pass@2	0.036	0.158	0.0004	0.053 / 0.034

The strongest claim I am comfortable making is:

On ARC-2, the LLM aggregate is materially closer to the human difficulty structure than the currently stored non-LLM systems are.

That said, there is a real nuance here. The best non-LLM profile is *not* the best-scoring TRM checkpoint. The more human-aligned TRM checkpoint is an earlier one.

5 Why The Low Correlations Are Still Informative

Figure 2 is the graph I most wanted after finishing the first pass. It answers the right skeptical question: “fine, but are these low correlations better than random placement at the same score level?”

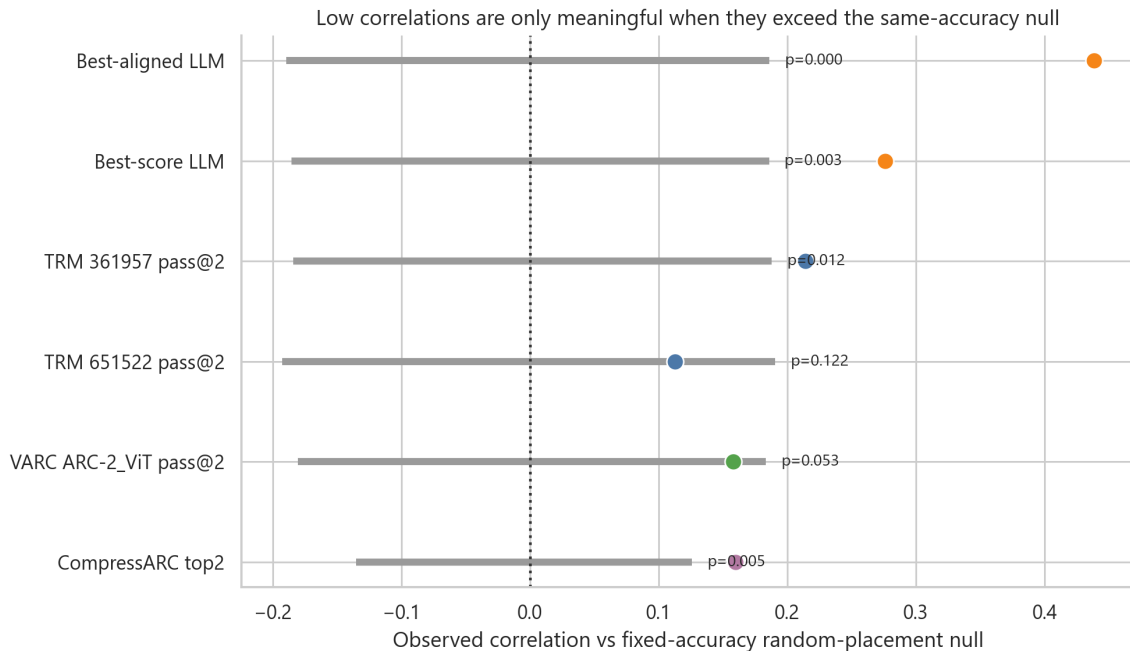


Figure 2: Observed correlations against the fixed-accuracy random-placement null. Gray bars show the 95% null interval given the same number of wins; colored points show the observed system correlation with human solve rates.

This changes the interpretation a lot:

- **TRM 361957** beats the same-accuracy null ($p = 0.012$).
- **VARC** is right on the edge ($p = 0.053$).
- **TRM 651522** does *not* beat the same-accuracy null ($p = 0.122$).

This is one of the clearest non-obvious findings in the whole note. The later TRM checkpoint is better at solving ARC items, but not better at solving the *human-like subset* of ARC items. Its extra wins appear to be placed in a less human-shaped way.

6 Residual Signal After Controls

The next question is whether the non-LLM systems are merely reacting to easy task features, or whether they retain a weaker version of the human difficulty axis even after simple controls.

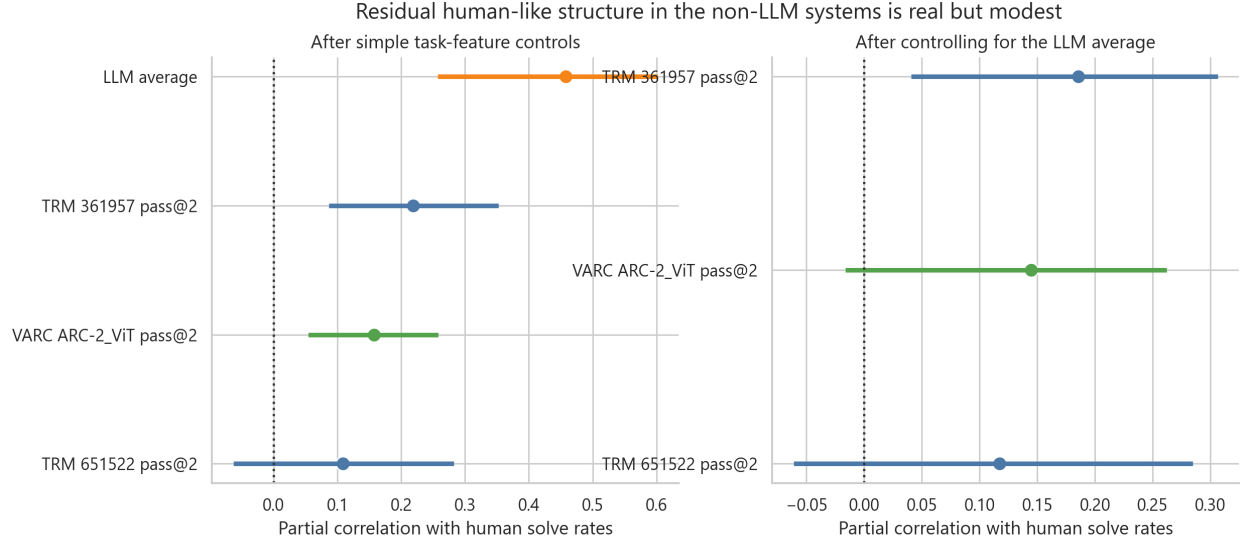


Figure 3: Residual human-like structure under two controls. Left: partial correlation after controlling for simple task features. Right: partial correlation after controlling for the LLM average.

There are really two stories here.

After simple task-feature controls

The feature-only null is too strong:

- LLM average: partial $r = 0.458$, $p < 0.001$
- TRM 361957: partial $r = 0.219$, $p = 0.001$
- VARC: partial $r = 0.157$, $p = 0.001$
- TRM 651522: partial $r = 0.108$, $p = 0.086$

So the observed alignment is not just “humans and models both like small tasks” or “they are both responding to color-count cues.”

After controlling for the LLM average

This is the tougher test:

- TRM 361957 keeps a modest residual: partial $r = 0.186$, $p = 0.009$, $q = 0.028$
- VARC is borderline: partial $r = 0.145$, $p = 0.034$, $q = 0.052$
- TRM 651522 does not keep a resolved residual: partial $r = 0.117$, $p = 0.112$

This is where I want to stay cautious. There is some residual non-LLM signal, especially for the mid-training TRM checkpoint. But it is small, and at least for VARC it is right on the multiple-testing boundary. So I would call this *real but modest*, not a dramatic separate human-like axis.

7 Complementarity and TRM Training Dynamics

The non-LLM systems are not redundant, even though they are less human-aligned overall. The strongest practical result here is complementarity: the TRM+VARC union rescues 6 human-easy / LLM-hard ARC-2 items, above the chance baseline ($p = 0.022$, $q = 0.037$).

That is enough to rule out the strongest version of “non-LLMs add nothing.” They do add something. It is just not the *same thing* as being globally closer to the human difficulty axis.

The TRM training curve makes the same point from another angle.

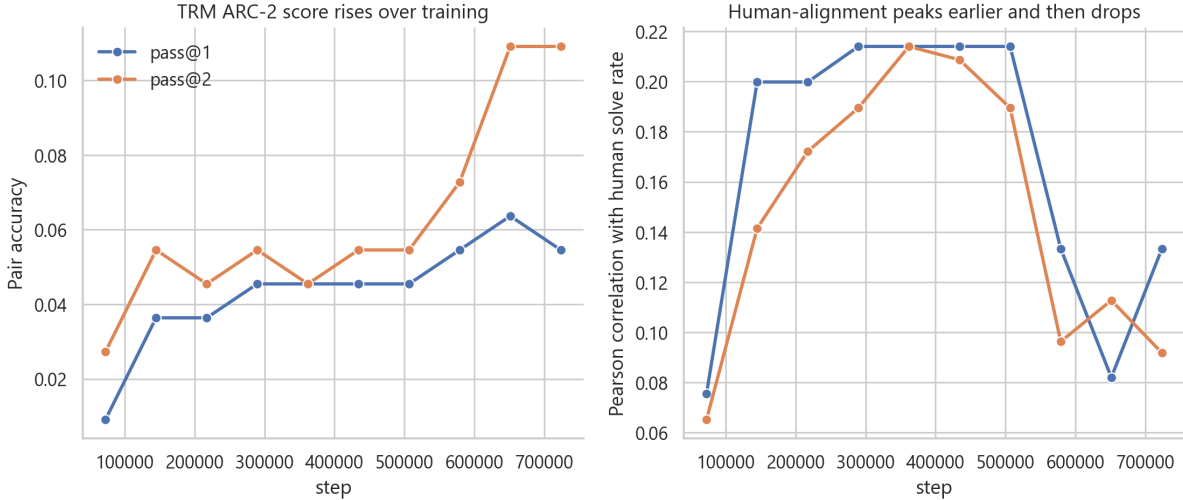


Figure 4: TRM training dynamics on the ARC-2 human-overlap subset. Accuracy rises over training, but human-alignment peaks earlier and does not rise in lockstep.

The formal trajectory tests are:

- step vs accuracy: Spearman $\rho = 0.850$, $p = 0.002$
- step vs human-correlation: Spearman $\rho = -0.085$, $p = 0.827$
- accuracy vs human-correlation: Spearman $\rho = -0.265$, $p = 0.451$

So training clearly improves benchmark score, but does not clearly improve human-likeness. If anything, the tendency goes the other way, though not significantly in this small checkpoint series.

8 ARC-1 Sidecar: CompressARC

There is still no dedicated ARC-1 human benchmark file in the repository, but there is a useful ARC-1 overlap through task reuse in the ARC-AGI-2 Public Train human data. That gives 230 single-pair ARC-1 evaluation tasks with a human split-half median of 0.372.

The ARC-1 sidecar result is:

- ARC1 LLM average: human $r = 0.285$, H1 $p = 0.057$
- ARC1 best-aligned LLM: human $r = 0.286$, H1 $p = 0.061$

- CompressARC top2: human $r = 0.160$, H1 $p < 0.001$
- CompressARC fixed-accuracy null: $p = 0.005$

So CompressARC is not random with respect to human difficulty. It is doing something real. But it is still not human-equivalent on this overlap, and it does not overturn the main ARC-2 picture.

9 What I Think This Means

Here is the most compact synthesis I trust:

1. We can reject the strong null that the current non-LLM systems are psychometrically indistinguishable from humans on ARC-2.
2. We should *not* say that the LLM aggregate is “proven the same as humans.” A high H1 p is a *fail-to-reject*, not a proof of equality. Still, the LLM aggregate clearly sits much closer to the human split-half benchmark than the non-LLM systems do.
3. We can reject the claim that all observed alignment is trivial feature matching.
4. We have some evidence for a weak non-LLM residual beyond the LLM average, especially for the mid-training TRM checkpoint, but it is not large enough to anchor a big philosophical claim.
5. We can reject the idea that non-LLMs add no complementary value at all.

So the cleanest overall statement is:

The LLM aggregate is currently the closest thing in this repository to the human ARC-2 difficulty axis, but the non-LLM systems still contribute complementary and partly human-relevant structure.

That is a more interesting result than either simplistic extreme. The non-LLM systems are neither just answer-key junk nor a secret hidden route to human-like ARC behavior. They look more like partial alternative solvers: sometimes useful, sometimes idiosyncratic, and not yet globally aligned with the human difficulty landscape.

10 Limits

The main limits are straightforward.

- The ARC-2 human subset is only 110 well-sampled test pairs.
- The non-LLM sample is small: one TRM trajectory, one VARC release, one CompressARC artifact.
- The current tests say nothing direct about *process*; they are about item-level difficulty profiles, not internal reasoning traces.
- The ARC-1 sidecar is useful, but it is still a sidecar rather than a dedicated ARC-1 human benchmark.

So this should be read as a careful repository note, not the final word.

A Extra Robustness Figures

I do not think these are the core figures, but I still wanted them nearby.

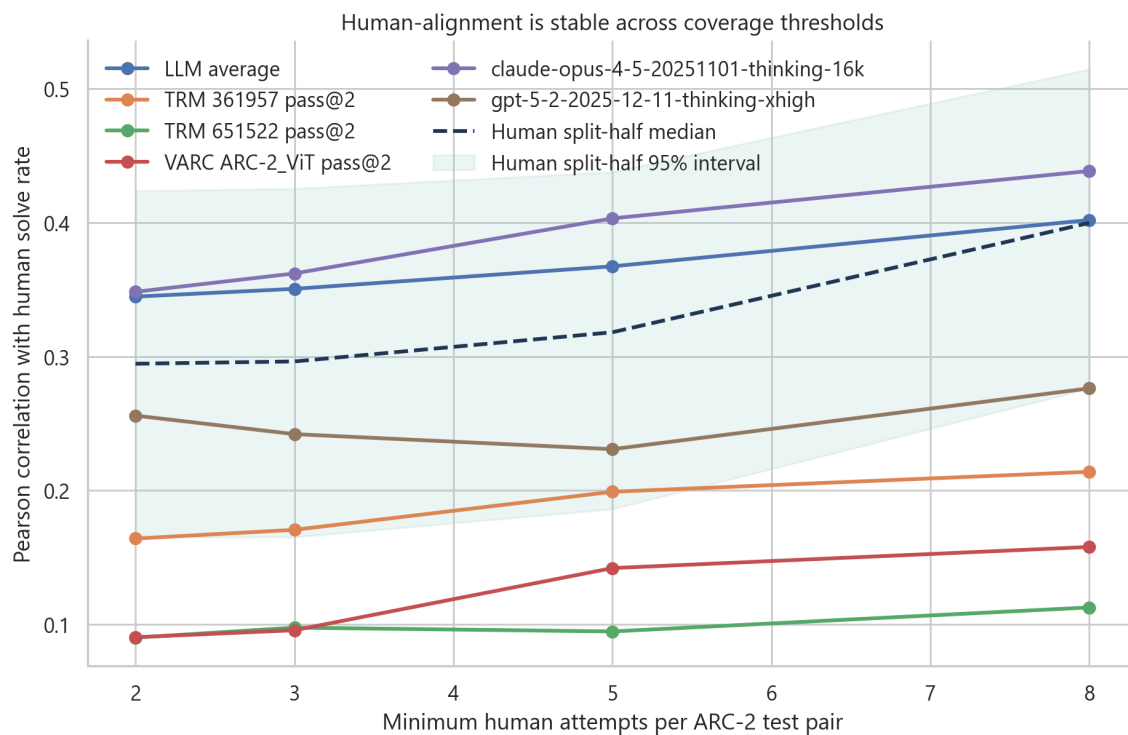


Figure 5: Threshold sensitivity. The main ranking is stable as the minimum human-attempt threshold changes.

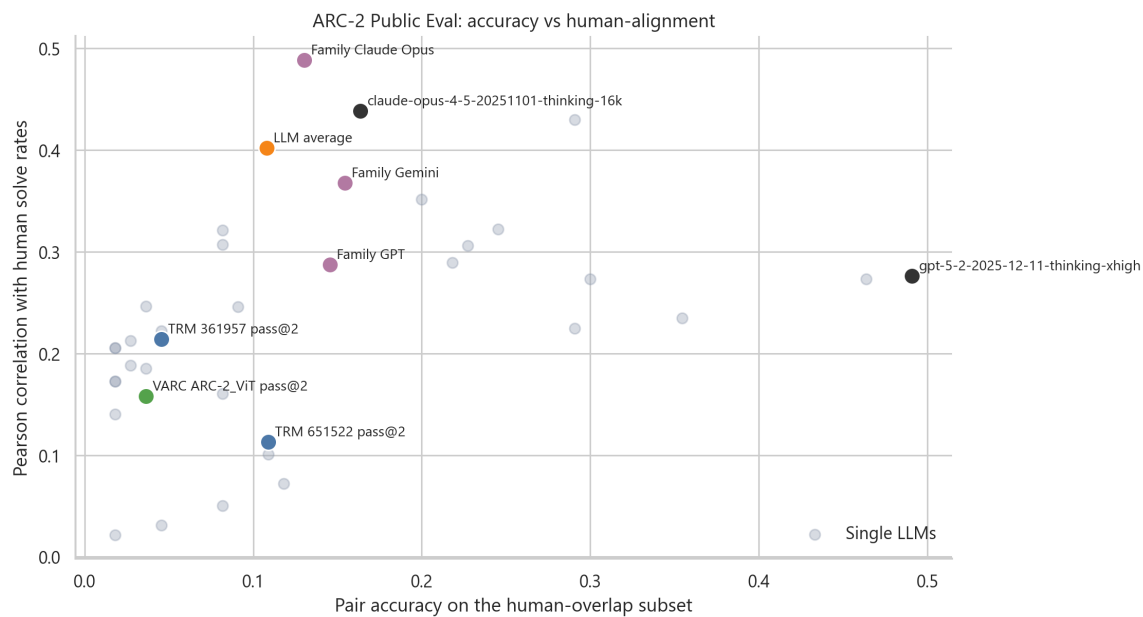


Figure 6: Accuracy versus human-alignment on the ARC-2 human-overlap subset. The main point here is that score and human-likeness are related but not interchangeable.